Two computable arrays of Turing indexes: a colouring from the k-th jump whose thin sets compute the k-th jump

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Editorial note. The counted unit is the article's own Lemma lem:arrays at source lines 683-685 together with its proof at source lines 687-697, a straight adjacent pairing in the pinned file. The pinned file is a concatenation of the authors' main and macro files; it is the article's fixed version arXiv:2412.11598v3 under CC BY 4.0.

Editorial note. This source ships no separate preamble file, so the retained preamble is sliced mechanically from the pinned file by the lane and is shared, byte for byte, with the other cards of this source; it carries the authors' full macro block so that no macro of a retained span is left undefined. The first card of this source needed one revision for exactly that reason.

Editorial note. The retained statement and proof contain no reference to another labelled result of the article and no citation, so this card carries no navigation pointer and no bibliography entry. The two arrays produced by the lemma are used by later results of the article, and those results are counted by other lanes; none of them is reproduced here.

Editorial note. The proof's final sentence is a uniformity claim about the computation it has just built, and it is part of the author's own argument rather than a pointer to another result, so it is retained verbatim.

Allowed standard prerequisites (editorial).

- Thin sets, uniform moduli of a jump and computation from a set. A set is thin for a colouring of finite tuples in a colour when no tuple of the set of the relevant length has that colour; a uniform modulus of a set is a function that dominates every function computed by that set; and a set computes another when the latter is decidable relative to the former, written with the usual Turing-reduction symbol. The author's own notation is retained inside the counted statement and proof.. The counted lemma is the engine of the section: it produces both the colouring lived on the k-th jump and the indexes that turn any infinite thin set back into the k-th jump, and the following results of the section are built on exactly those two arrays.

Lemma 0.1. *There exists two computable arrays $(e_{n,k})_{n,k\in\mathbb{N}}$ and $(d_{n,k})_{n,k\in\mathbb{N}}$ of Turing indexes such that for every $n, k \in \mathbb{N}^+$ with $n \geq 2$, $\Phi^{\emptyset^{(k)}}_{e_{n,k}}$ is a coloring $f^k_n : [\mathbb{N}]^n \to n$ such that for every infinite f^k_n-thin set H, $\Phi^H_{d_{n,k}} = \emptyset^{(k)}$.*

Proof. Let $k \in \mathbb{N}^+$. Let g_k be a uniform modulus of the set $\emptyset^{(k)}$. Note that Δ^0_{k+1}-indexes for each function g_k can be found computably, uniformly in k.

For $n \geq 2$ let $f^k_n(x_0, \dots, x_{n-1}) = n-1$ if $g_k(x_0) \leq x_1$ and if $g_k(x_0) > x_1$, let $f^k_n(x_0, \dots, x_{n-1}) = i$ for $i < n-1$ the largest value such that $g_k(x_0) > x_{i+1}$.

Let H be an infinite f^k_n-thin set for some color c. The color c cannot be $n-1$ as for a given $x_0 \in H$ there exists $x_1 < \cdots < x_{n-1} \in H \setminus \{0, \dots, g_k(x_0)\}$, hence $f^k_n(x_0, \dots, x_{n-1}) = n-1$.

We claim H is thin for color $n-2$. Indeed, assume by contradiction that there exists some tuple $x_0 < \cdots < x_{n-1} \in H$ such that $g_k(x_0) > x_{n-1}$ so that f^k_n takes color $n-2$ on $\{x_0, \dots, x_{n-1}\}$. Since $c < n-2$ and H is infinite, there exists some $y_{c+2} < \cdots < y_{n-1} \in H \setminus \{0, \dots, g_k(x_0)\}$, therefore $f^k_n(x_0, \dots, x_{c+1}, y_{c+2}, \dots, y_{n-1}) = c$, contradicting our assumption that H is f^k_n-thin for c.

Write $H = \{x_0 < x_1 < \dots\}$. For every $i \in \mathbb{N}$, $g_k(x_i) \leq x_{i+n-1}$, hence H computes a function dominating g_k and therefore computes $\emptyset^{(k)}$. This computation can be done uniformly in the set H, k and n. □